Cross-dataset fairness tables in the FairForensics (Table III) format: current status

CES-FL project, 6 Nov 2026. Aggregates of existing saved scores; no new inference.

Format. As FairForensics (arXiv 2608.01661) Table III: per test set, F_{FPR}, F_{MEO}, F_{DP}, F_{OAE} over 8 intersectional demographic subgroups (sex × age 0–29/30+ × White/Non-White, labelled with the official FairFace classifier, as in their protocol) and AUC, all in %. **Table 1 (filled):** train on FF++ (Ding protocol), test on FF++ and four unseen datasets including FairForge-2. **Table 2 (empty):** the exact FairForensics analogue, train on FairForge-2 TRAIN and test on external datasets; no FF2-trained model has been evaluated there yet. Threshold: each model's FF++-validation rule (A1). Video sets use video-majority labels; a group enters FPR terms only with ≥3 real videos (persons on FF2). Values are seed means (seed count shown). These are descriptive; no significance claims. Bold = best per column among rows shown.

Table 1: Performance comparison under cross-dataset evaluation, trained on FF++ (Ding et al. protocol). Detection and fairness (%) on FF++ (in-domain), Celeb-DF, DFDC, DFD and FairForge-2 public TEST (87,564 images; every FF2 generator is unseen for these models). Lower fairness scores are better; higher AUC is better.

Type	Detector	Seeds	FF++ (in-domain)					Celeb-DF					DFDC					DFD					FairForge-2 TEST				
			$F_{\mathrm{FPR}}\downarrow$	$F_{\mathrm{MEO}}\downarrow$	$F_{\mathrm{DP}}\downarrow$	$F_{\mathrm{OAE}}\downarrow$	AUC↑	$F_{\mathrm{FPR}}\downarrow$	$F_{\mathrm{MEO}}\downarrow$	$F_{\mathrm{DP}}\downarrow$	$F_{\mathrm{OAE}}\downarrow$	AUC↑	$F_{\mathrm{FPR}}\downarrow$	$F_{\mathrm{MEO}}\downarrow$	$F_{\mathrm{DP}}\downarrow$	$F_{\mathrm{OAE}}\downarrow$	AUC↑	$F_{\mathrm{FPR}}\downarrow$	$F_{\mathrm{MEO}}\downarrow$	$F_{\mathrm{DP}}\downarrow$	$F_{\mathrm{OAE}}\downarrow$	AUC↑	$F_{\mathrm{FPR}}\downarrow$	$F_{\mathrm{MEO}}\downarrow$	$F_{\mathrm{DP}}\downarrow$	$F_{\mathrm{OAE}}\downarrow$	AUC↑
Fair-based Xception	Ding et al. CVPR'26 Ori (our repro.)	3	18.54	34.74	40.93	18.97	94.11	56.58	46.57	36.13	20.68	67.14	40.88	28.49	13.90	27.69	54.68	51.10	47.64	25.37	29.93	66.80	44.02	19.56	14.48	12.46	37.90
	Ding et al. CVPR'26 SFD+GDA (our repro.)	1	7.60	36.63	45.62	17.08	97.44	59.36	57.33	46.00	21.37	71.22	18.20	19.20	6.15	30.07	56.05	51.40	43.37	24.45	28.75	72.54	41.75	18.12	13.53	11.68	36.36
Ours (CFES) CLIP ViT-L/14	Detection-only control	3	46.53	30.34	27.52	19.25	91.55	84.17	61.98	48.99	24.39	81.41	42.66	30.30	16.51	**17.14**	71.75	34.25	37.23	17.55	16.24	83.68	36.02	17.60	15.62	15.29	62.71
	CFES A (rev4)	3	50.35	30.78	**24.82**	22.27	92.99	85.81	58.55	45.20	26.91	85.66	37.85	30.83	15.55	17.44	73.22	37.60	**33.22**	20.19	14.79	86.97	24.56	15.58	14.68	14.62	66.64
	arm C	3	30.37	32.93	32.92	23.66	95.80	36.95	38.56	40.13	21.08	**89.06**	21.26	37.18	9.86	21.82	75.09	33.06	34.79	18.84	10.47	**90.68**	24.99	19.08	17.94	17.93	69.90
	arm E w1	1	13.32	51.48	50.88	28.81	97.03	**28.12**	49.05	44.29	28.82	87.56	**5.99**	30.05	8.73	26.35	**79.19**	**9.20**	66.63	35.05	25.20	89.66	**6.78**	**13.44**	12.33	12.91	76.92
	arm E w3	1	8.68	52.99	52.52	28.20	97.06	30.96	37.37	40.13	22.69	86.87	6.08	29.76	7.32	27.13	77.92	10.12	65.39	33.14	19.55	89.72	7.64	14.38	13.16	14.11	78.46
	arm E w10	1	10.66	53.62	50.19	29.38	96.67	35.69	**22.86**	**19.25**	**13.35**	84.89	8.12	33.88	8.55	26.80	76.41	14.42	64.07	25.98	**10.10**	90.05	16.11	19.07	17.59	18.36	**79.22**
	CFES-D (final)	3	24.68	37.84	37.89	23.86	95.61	32.69	33.85	33.50	20.93	87.31	14.92	33.06	8.71	21.80	76.92	50.02	45.83	19.88	16.60	89.90	25.14	18.26	13.66	13.03	70.99
Ours (CFES) ResNet-50	CFES-ResNet	3	36.78	**25.92**	34.14	**15.95**	92.52	53.49	40.73	30.75	19.83	71.38	35.16	23.07	13.51	21.49	58.66	51.72	45.41	**16.77**	22.21	71.40	37.68	17.62	**11.41**	**10.33**	38.48
Ours (CES-FL) Xception	L_A (plain head, BCE)	1	5.42	46.88	46.54	24.04	98.47	49.33	36.23	31.99	13.88	70.75	10.76	**13.08**	5.49	27.98	61.77	41.87	43.80	23.03	26.43	81.16	– not evaluated –				
	L_H (plain head, ours)	1	**3.04**	48.96	47.64	24.91	**98.49**	39.39	32.92	27.44	16.34	71.40	7.54	13.70	**4.86**	28.55	60.46	42.76	41.98	21.35	23.04	80.88	– not evaluated –				

Reading Table 1. On the FairForge-2 column the arm E variants reach 77–79 AUC (one seed each) and CFES-D 70.99 (3 seeds), against 36–38 for the Ding reproductions. FairForge-2 is hard for every FF++-trained detector, and Ding's method transfers worst. The CES-FL L_A/L_H rows (Xception, plain head, seed 17) were rescored on these 8 groups from their saved 3 Nov scores (no new inference), with the A1 threshold re-derived on FF++ validation; they have no FairForge-2 TEST scores. At A1 their DFDC TPR is only about 10%, so the low DFDC fairness gaps there reflect an almost-never-fake operating point.

Table 2: The exact FairForensics analogue: train on FairForge-2 public TRAIN (218,266 images), test on unseen datasets. **Status: not evaluated.** Checkpoints marked "done (VAL)" exist and are verified on FF2 validation; none has been run on the external test sets. That one-time evaluation (inference only, frozen checkpoints and thresholds) needs approval.

Detector (trained on FF2)	FF2 checkpoint	FF++					Celeb-DF					DFDC					DFD				
		F_{FPR}	F_{MEO}	F_{DP}	F_{OAE}	AUC	F_{FPR}	F_{MEO}	F_{DP}	F_{OAE}	AUC	F_{FPR}	F_{MEO}	F_{DP}	F_{OAE}	AUC	F_{FPR}	F_{MEO}	F_{DP}	F_{OAE}	AUC
Ori (BCE)	done (VAL)	– not evaluated –																			
DAG-FDD	done (VAL)	– not evaluated –																			
DAW-FDD	done (VAL)	– not evaluated –																			
PG-FDD	not trained	– not evaluated –																			
F3Net (FAD)	trained; verification failed	– not evaluated –																			
CORE	trained; verification failed	– not evaluated –																			
Ding et al. (corrected GDA, our run)	done (VAL)	– not evaluated –																			
CES-FL P	done (VAL)	– not evaluated –																			
CES-FL A	done (VAL)	– not evaluated –																			
CES-FL B	done (VAL)	– not evaluated –																			
L_A (matched control)	done (VAL)	– not evaluated –																			
L_H-PS (ours)	awaiting probe review	– not evaluated –																			